

Computational Study of TEMPO Radical as a Potential Qubit Material

Intermediate MTP Report

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1 Abstract

This project examines the computational evaluation of spin-related properties in molecular systems, with a focus on the impact of vibrational modes on electronic parameters. Using quantum mechanical methods, we calculate key quantities such as g-tensors and coupling parameters. It demonstrates how vibrational coupling influences spin dynamics, highlighting potential strategies to optimize qubit stability. The results contribute to understanding spin-phonon interactions and their role in quantum information applications.

2 Introduction

To make practical quantum technologies viable, it is essential to prepare, protect, and manipulate coherent quantum states over sufficiently long timescales. Molecular qubits, although promising, are particularly susceptible to *decoherence* because of their unavoidable interactions with the surrounding environment. Processes such as molecular vibrations, spin-spin interactions, spin-lattice mechanisms, and even external machinery noise collectively act to destroy phase coherence. As a result, maintaining and preserving quantum information with high accuracy remains a considerable challenge. Achieving long coherence times, especially under room-temperature conditions, remains one of the central obstacles in the development of quantum materials and devices.

Spin-based molecular qubits have emerged as an interesting field because their chemical structures can be manipulated with atomic precision, and they can be incorporated into diverse solid-state architectures. Organic radicals, particularly nitroxide radicals such as 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO), have therefore attracted extensive interest. Their unpaired electron, localised primarily on the N-O moiety, generates a well-defined two-level spin system that can serve as a qubit. Furthermore, the surrounding molecular framework can be chemically engineered to reduce environmental coupling, suppress decoherence pathways, and enhance relaxation times.

Understanding how the structural and environmental factors influence coherence is essential for transforming these molecular spin probes into functional quantum units. Studies on TEMPO provide a valuable starting point for exploring how quantum coherence emerges from, and is influenced by, molecular design, nuclear spin environments, and vibrational dynamics. Such knowledge directly informs the rational design of next-generation molecular qubits, which exhibit improved performance and stability.

A significant challenge in the field has been integrating molecular spin qubits into surface-bound architectures without compromising coherence. Traditional deposition methods often led to chemical instability, surface-induced decoherence, or loss of spin integrity. Recently, however, a bottom-up self-assembled monolayer (SAM) strategy has been demonstrated, enabling the formation of robust surface-anchored molecular qubits. Remarkably, this SAM-based approach not only preserves the intrinsic coherence properties but has been shown to enhance coherence times to values as high as $13\mu\text{s}$, highlighting the potential of molecular engineering in pushing the limits of quantum coherence.

3 Objectives

- Develop a model to investigate the key mechanisms responsible for rapid decoherence in molecular spin systems.
- Using EPR spectroscopy and simulations to quantify relaxation parameters.

- Determine the coupling parameters in such processes.

4 Computational Methodology

All electronic structure calculations were carried out using Density Functional Theory (DFT). Geometry optimizations for all molecular systems were performed using *Gaussian* software package. The structures were optimized without imposing any symmetry constraints, allowing the system to explore the full configurational space. The B3LYP functional and def2-TZVP basis set was used, as it provides a balanced triple-zeta description with appropriate polarization functions and is regarded as a reliable choice for geometry optimization.

For species with unpaired electrons, as in the case of TEMPO radical, an *open-shell* electronic structure treatment was used. In particular, the Unrestricted Kohn–Sham (UKS) formalism ensures a physically meaningful description of the spin distribution and allows for the proper treatment of the spin density, which is essential for subsequent property calculations.

Electron paramagnetic resonance (EPR) properties were computed using the *ORCA* quantum chemistry package. The optimized geometries obtained from *Gaussian* were used directly, with no further structural refinement. Computation of *g*-tensors and hyperfine coupling tensors was performed using *ORCA*. The same basis was retained for all atoms to maintain methodological consistency across all stages of the calculation.

The resulting magnetic parameters, including the isotropic and anisotropic components of the *g*-tensor and the full hyperfine tensor for relevant nuclei were extracted from the log files. These theoretical quantities provide important insight into the electronic distribution, local spin environment, and the nature of the unpaired electron in the studied systems. They form the basis for direct comparison with experiments. To gain further insights into the coupling parameters, a home-designed *Python* program was used, which enabled fitting of the data and determination of the required parameters.

For the simulation and visualization of EPR spectra, *EasySpin* in *matlab* was used. *EasySpin* offers a range of simulation modules for different experimental conditions. In this work, the *pepper* function was used, as it is specifically designed for continuous-wave EPR experiments in the solid state, making it suitable for simulating the TEMPO radical under frozen-solution or solid-state conditions. The simulation parameters, such as temperature, microwave frequency, magnetic field strength, and the desired magnetic field sweep range, can be specified by the user. In this work, a field range of 325–350 mT was used to generate the simulated spectra.

5 Results & Discussion

5.1 Hamiltonian Construction

Continuous-wave and pulsed EPR measurements confirm that TEMPO possesses a stable $S = 1/2$, with the unpaired electron primarily present over the nitroxide moiety. This delocalisation reduces the hyperfine anisotropy and yields a well-defined *g*-tensor, both of which are advantageous for coherent qubit manipulation. In the absence of external magnetic field, $\vec{B}$, the two spin states $|0\rangle$ and $|1\rangle$ are degenerate when there is a single spin centre with no interaction with the neighbours. When the magnetic field is now switched on, the states split (Zeeman splitting) and the electron gets into the excited state. Turning the magnetic field on only separates the two spin levels, and turning it off

makes them degenerate again. Decoherence is simply the loss of phase coherence, not a return to the ground state.

Now, we take a molecular system containing multiple spin centers. Now, this can interact with the other spin centers, leading to a complex relaxation process, which includes spin-spin, spin-lattice, molecular vibrations, and machinery disturbances. A complete molecular Hamiltonian includes contributions from the electronic orbital motion, electronic spin, nuclear motion (rotation and vibration), nuclear spin, and all associated coupling terms such as spin-orbit, hyperfine, spin-spin, and spin-rotation interactions. [1]

$$\hat{H} = \hat{H}_{\text{elec}} + \hat{H}_{\text{nuc}} + \hat{H}_{\text{vib}} + \hat{H}_{\text{rot}} + \hat{H}_{\text{SO}} + \hat{H}_{\text{SS}} + \hat{H}_{\text{SR}} + \hat{H}_{\text{HF}} \quad (1)$$

A magnetic moment $\boldsymbol{\mu}$ placed in an external magnetic field $\mathbf{B}$ has an interaction energy

$$E_{\text{int}} = -\boldsymbol{\mu} \cdot \mathbf{B}. \quad (2)$$

Expressing via the components in the vectors in the Cartesian system,

$$E_{\text{int}} = -(\mu_x B_x + \mu_y B_y + \mu_z B_z). \quad (3)$$

In quantum mechanics, the magnetic moment becomes an operator $\hat{\boldsymbol{\mu}}$, giving the field-dependent part of the Hamiltonian

$$\hat{H}_{\text{field}} = -\hat{\boldsymbol{\mu}} \cdot \mathbf{B} = -(B_x \hat{\mu}_x + B_y \hat{\mu}_y + B_z \hat{\mu}_z) = -\mathbf{B}^T \hat{\boldsymbol{\mu}}. \quad (4)$$

Let $\hat{H}^0$ denote the field-independent contributions (zero-field splitting, hyperfine, nuclear quadrupole, electron-electron interactions, etc.). The full spin Hamiltonian in an applied field is therefore

$$\hat{H}(\mathbf{B}) = \hat{H}^0 - \mathbf{B}^T \hat{\boldsymbol{\mu}}. \quad (5)$$

The total magnetic-moment operator consists of electron and nuclear parts,

$$\hat{\boldsymbol{\mu}} = \hat{\boldsymbol{\mu}}^{(e)} + \hat{\boldsymbol{\mu}}^{(n)}. \quad (6)$$

For electrons and nuclei,

$$\hat{\boldsymbol{\mu}}^{(e)} = -\mu_B \mathbf{g} \hat{\mathbf{S}}, \quad (7)$$

$$\hat{\boldsymbol{\mu}}^{(n)} = +g_n \mu_N \hat{\mathbf{I}}, \quad (8)$$

where μ_B and μ_N are the electron Bohr and nuclear Bohr magnetons respectively, $\mathbf{g}$ is the electron g -tensor, and g_n is the nuclear g -factor.

Thus the electron Zeeman term is

$$\hat{H}_{\text{EZ}} = -\mathbf{B}^T \hat{\boldsymbol{\mu}}^{(e)} = \mu_B \mathbf{B}^T \mathbf{g} \hat{\mathbf{S}}, \quad (9)$$

and the nuclear Zeeman term is

$$\hat{H}_{\text{NZ}} = -\mathbf{B}^T \hat{\boldsymbol{\mu}}^{(n)} = -g_n \mu_N \mathbf{B}^T \hat{\mathbf{I}}. \quad (10)$$

The total electron-nuclear magnetic interaction consists of two main contributions: the isotropic Fermi contact term and the anisotropic magnetic dipole-dipole interaction. These contributions are collected into the hyperfine tensor $\mathbf{A}$. The hyperfine Hamiltonian can therefore be written as

$$\hat{H}_{\text{hf}} = \hat{\mathbf{S}}^T \mathbf{A} \hat{\mathbf{I}}, \quad (11)$$

where $\hat{S}$ is the electron spin operator.

If we only consider the electronic Zeeman part and the hyperfine interaction, the Hamiltonian simplifies to,

$$\hat{H}_{\text{spin}} = \mu_B \mathbf{B}^T \mathbf{g} \hat{S} + \hat{S}^T \mathbf{A} \hat{I}. \quad (12)$$

where B is the applied magnetic field (a vector), $\hat{S}$ and $\hat{I}$ are the electron and nuclear spin operators, respectively, and g is the electron g -tensor.

In terms of the direction cosines and individual components of g and A , the following expressions for the *effective* g and A values can be obtained.

$$g_{\text{eff}}(\alpha, \beta) = \sqrt{g_{xx}^2 \cos^2 \alpha \sin^2 \beta + g_{yy}^2 \sin^2 \alpha \sin^2 \beta + g_{zz}^2 \cos^2 \beta} \quad (13)$$

$$A_{\text{eff}}(\alpha, \beta) = \sqrt{A_{xx}^2 \cos^2 \alpha \sin^2 \beta + A_{yy}^2 \sin^2 \alpha \sin^2 \beta + A_{zz}^2 \cos^2 \beta} \quad (14)$$

The angles α and β define the orientation of the external magnetic field $\vec{B}$ relative to the molecular frame (x, y, z). Using spherical coordinates in the molecular frame, we have:

$$B_x = B \sin \beta \cos \alpha, \quad B_y = B \sin \beta \sin \alpha, \quad B_z = B \cos \beta \quad (15)$$

Here, α is the azimuthal angle (rotation around the molecular z-axis) and β is the polar angle (angle from the molecular z-axis).

5.2 Spin Density & the Frontier MOs

The spin density shows that the unpaired electron is primarily delocalized across the nitroxide N-O bond. This N-O moiety is confirmed as the primary site for radical reactions. The bulky methyl groups on both sides provide steric shielding without significantly participating in the electronic radical character.

As an open-shell system, TEMPO's reactivity is governed by its Singly Occupied Molecular Orbital (SOMO) and the Lowest Unoccupied Molecular Orbital (LUMO). (Refer to Figure 1)

5.3 Normal Modes of Vibration of TEMPO

In our study of TEMPO radicals, we obtained the frequencies of the normal modes from ab initio calculations in gaussian. As suggested by Debye theory, not all modes are active at a certain temperature. At sufficiently low temperature (10 K), under the harmonic oscillator approximation, equating $h\nu/2$ to thermal energy at that temperature gives $\nu = 4.17 \times 10^{11} \text{ s}^{-1}$ or 13.88 cm^{-1} . This gives us a rough idea that modes below 13.88 cm^{-1} are vibrationally active at 10 K. However, the frequency starts from 64.17 cm^{-1} . That means, no mode will be active at 10 K. However, we took the first six normal modes (Refer to Table 1 and Figure 2) for our analysis and investigate their effects, such as the variation of the components of the 'g tensor' and the 'hyperfine matrix' along the normal modes, calculating their coupling parameters and so on.

For small enough displacements, the Taylor expansion of a molecular property g_k along normal mode at absolute temperature is given by [2]:

$$g(Q_k) = g(0) + \left[\frac{\partial g_k}{\partial Q_k} \right]_{Q_k=0} Q_k + \frac{1}{2} \left[\frac{\partial^2 g_k}{\partial Q_k^2} \right]_{Q_k=0} Q_k^2 + \dots \quad (16)$$

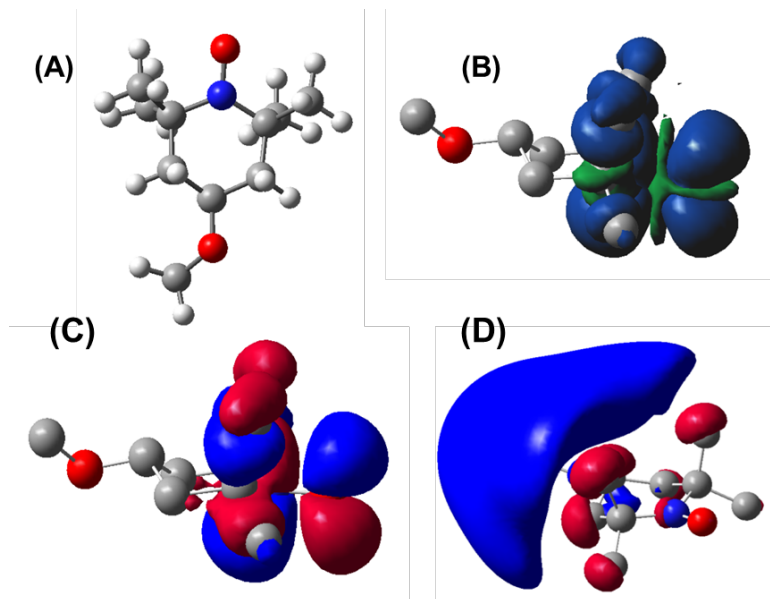


Figure 1: (A): *TEMPO optimized structure*; (B): *Spin density of TEMPO* (C), (D): *HOMO & LUMO of TEMPO, respectively*

Here, $g(0)$ is the value of the property at the equilibrium. The first derivative term, $\left[\frac{\partial g_k}{\partial Q_k}\right]_{Q_k=0} Q_k$, represents the linear change of g_k with the displacement along the normal mode and quantifies the strength of vibronic coupling. The second derivative term, $\frac{1}{2} \left[\frac{\partial^2 g_k}{\partial Q_k^2}\right]_{Q_k=0} Q_k^2$, accounts for the non-linear contribution of the displacement.

Table 1: *Normal modes frequency values for modes 1 to 6*

# Normal Mode	Frequency (cm ⁻¹)
1	64.1654
2	77.1894
3	115.7038
4	123.3583
5	175.1996
6	196.0182

Now, we have distorted the optimized TEMPO radical from its equilibrium geometry from -1 to +1 dimensionless displacement only for the first six normal modes, and calculate their *g-tensor*, *A matrix* in *orca* and then fitted the data to a python code that determines the zeroth order, first order and second order coupling parameters of the individual components of *g*, as well as the isotropic *g*. Based on the value of the coupling parameters, one can decide the dominant term (Refer to Table 2).

The zeroth order coupling values for the respective component of the *g* tensor comes out to be the same as expected ($g_{xx} \approx 2.008925$, $g_{yy} \approx 2.005314$, $g_{zz} \approx 2.003472$), because it arises from the equilibrium geometry of the molecule, which comes after optimization of the molecule.

Now that we have considered anisotropy to obtain the individual diagonal components of the *g-matrix*, our next task was to get the isotropic *g* value and do a similar fitting to obtain the zeroth, first, and second order coupling parameters (Refer to Figure 4). The results are summarized in Table 3.

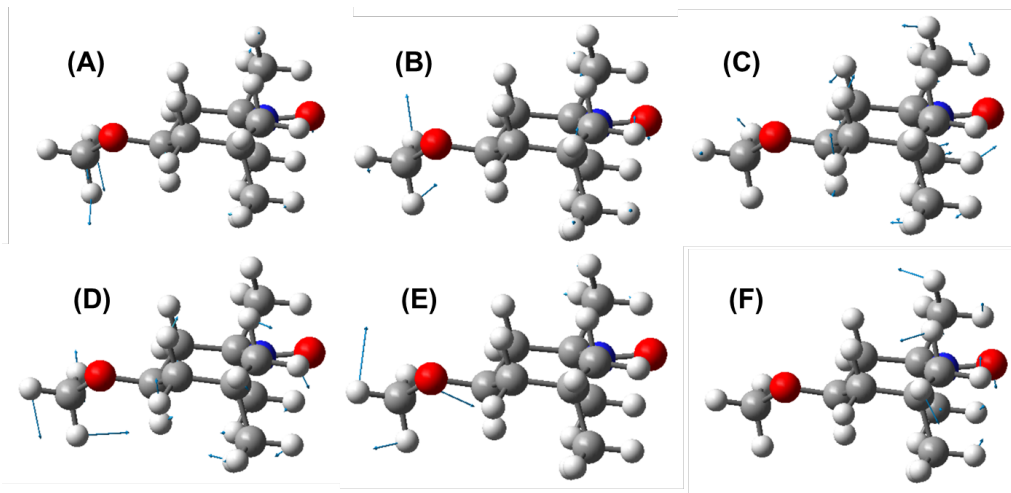


Figure 2: (A) to (F): Normal modes of vibration for modes 1 to 6, respectively

Similar fitting was done for the *hyperfine A matrix*, and the nature of the plots was studied (Refer to Figure 5).

It is evident from the values of the coupling parameters that mode 6 is the most significant contributor to the change in g and A tensors. The canonical ensemble probability analysis indicates that the population of vibrational mode 6 at 10 K is vanishingly small, in the order of 5.6 parts per billion. Despite this negligible thermal occupation, mode 6 emerges as the most significant contributor to the modulation of the coupling parameters. We believe this behaviour arises from the specific atomic displacements associated with mode 6, which predominantly move the atoms located at and immediately adjacent to the spin centre of the TEMPO radical (the N–O bond and the methyl groups on its two sides). Because the electron density is concentrated in that region, even a minute vibrational amplitude is able to produce substantial variations in both the g -tensor and the hyperfine A -tensor. Thus, the influence of mode 6 on the spin-phonon coupling does not arise from its thermal population but rather from the strong sensitivity of this mode. Our findings demonstrate that mode 6 plays a dominant role in determining the coupling parameters, whereas other modes with higher thermal populations contribute far less unless their nuclear motions directly perturb the atoms at the spin centre. *It is demonstrated that thermal occupation alone is therefore not a reliable indicator of the importance of a vibrational mode. The decisive factor is whether the mode involves geometrical changes that significantly affect the local environment of the unpaired electron.*

The same type of study was performed, taking two and three TEMPO radicals, separated by $\approx 5\text{ \AA}$ and their EPR spectra were studied (Refer to Figure 6). The EPR spectrum for a pair of TEMPO radicals exhibits features arising from some constructive and destructive overlap of their individual signals. These interference effects average out the characteristic hyperfine structure, causing the overall spectrum to appear more singlet-like.

5.4 Spin-Phonon Coupling

Spin-phonon coupling describes the interaction between the electronic spin of a paramagnetic centre and the vibrational motions (phonons) of the molecule or lattice, which indeed is responsible for the relaxation time.

The first-order coupling corresponds to the linear change of a spin Hamiltonian parameter (e.g.,

g -tensor or hyperfine tensor A) with respect to a normal mode coordinate. A large first-order coupling indicates that the particular vibrational mode k strongly modulates the Spin Hamiltonian, leading to more efficient relaxation or decoherence via this mode.

The second-order coupling corresponds to the quadratic change of the spin Hamiltonian parameter. Second-order terms account for the indirect or weaker modulation of spin parameters, including contributions from virtual excitations of higher electronic states.

Larger derivatives (first or second order) indicate stronger coupling of the spin to that vibrational mode, contributing more to relaxation and decoherence. The sign indicates whether the mode increases or decreases the spin parameter as the nuclei are displaced along that mode.

Table 2: Normal mode g -tensor components and couplings (dimensionless) for first 6 modes

# Normal Mode	Component	0 th Order	1 st Order ($\times 10^{-6}$)	2 nd Order ($\times 10^{-6}$)
1	g_{xx}	2.008925	235.9181	2.604895
	g_{yy}	2.005314	-1.140909	6.279137
	g_{zz}	2.003472	-168.1909	13.73543
2	g_{xx}	2.008925	202.8090	2.587412
	g_{yy}	2.005314	5.845454	7.261072
	g_{zz}	2.003472	-162.6000	9.702797
3	g_{xx}	2.008925	-14.74545	-2.663170
	g_{yy}	2.005314	3.309090	-0.559441
	g_{zz}	2.003472	-22.05909	-1.354895
4	g_{xx}	2.008924	33.19545	0.574009
	g_{yy}	2.005314	1.022727	-0.661422
	g_{zz}	2.003472	18.35000	0.043706
5	g_{xx}	2.008925	6.859090	2.572843
	g_{yy}	2.005314	7.786363	1.022727
	g_{zz}	2.003472	30.85909	0.731352
6	g_{xx}	2.008924	446.8227	66.17424
	g_{yy}	2.005314	6.604545	25.97028
	g_{zz}	2.003473	-230.7000	27.42424

Table 3: Isotropic g -tensor and their couplings (dimensionless) for the first 6 modes

# Normal Mode	0 th Order	1 st Order ($\times 10^{-6}$)	2 nd Order ($\times 10^{-6}$)
1	2.005904	20.05904	7.552447
2	2.005904	15.37273	6.509324
3	2.005904	-11.15455	-1.456877
4	2.005904	17.56818	0.014569
5	2.005904	15.15455	1.486014
6	2.005904	74.25000	3.987471

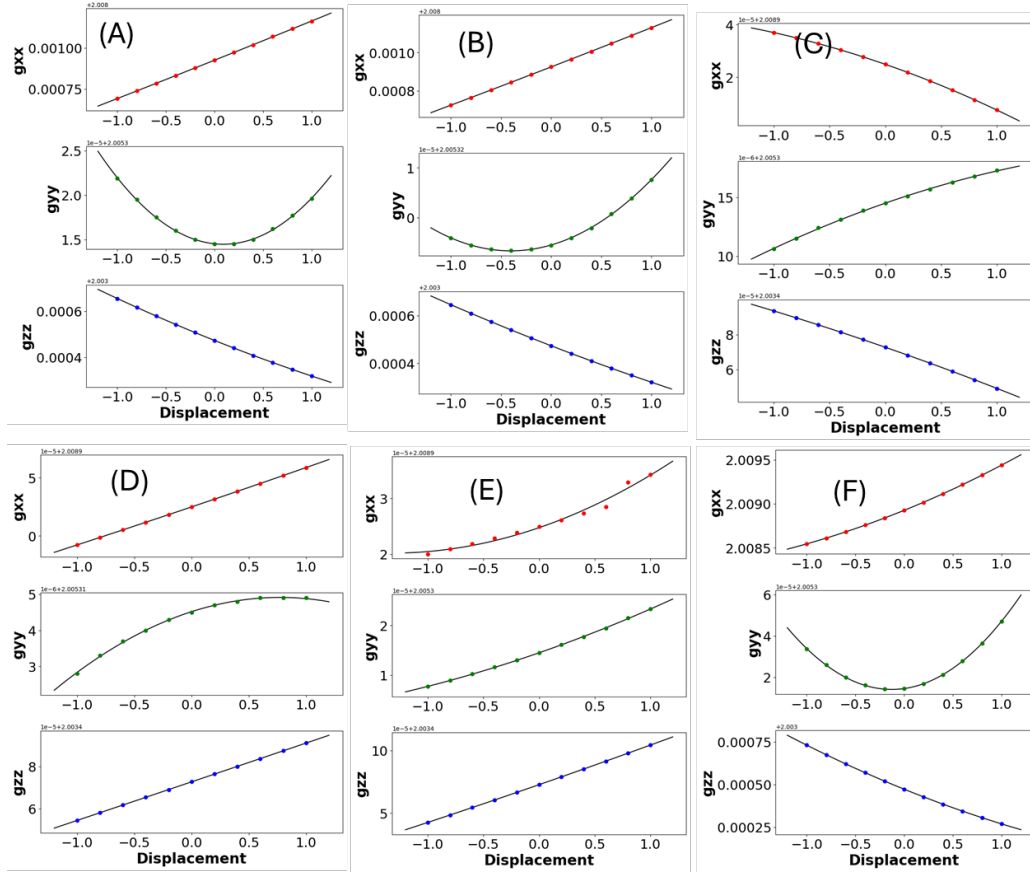


Figure 3: (A) to (F): Fitting of g tensor components to obtain coupling parameters for modes 1 to 6, respectively

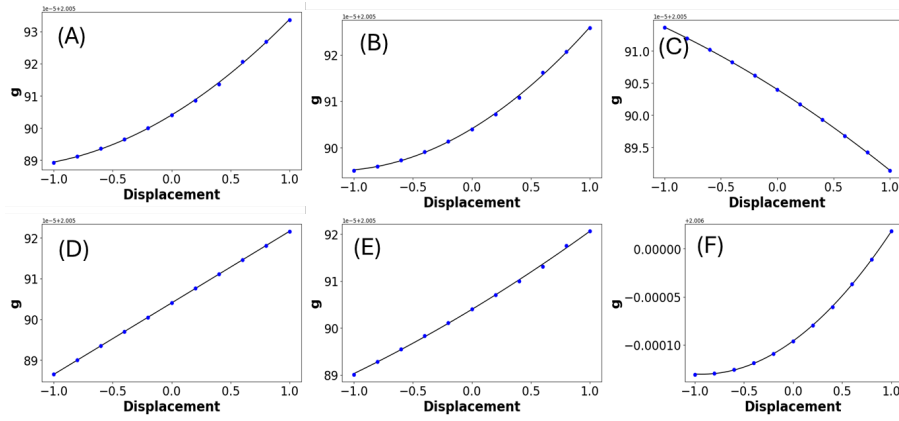


Figure 4: (A) to (F): Fitting of g_{iso} to obtain coupling parameters for modes 1 to 6, respectively

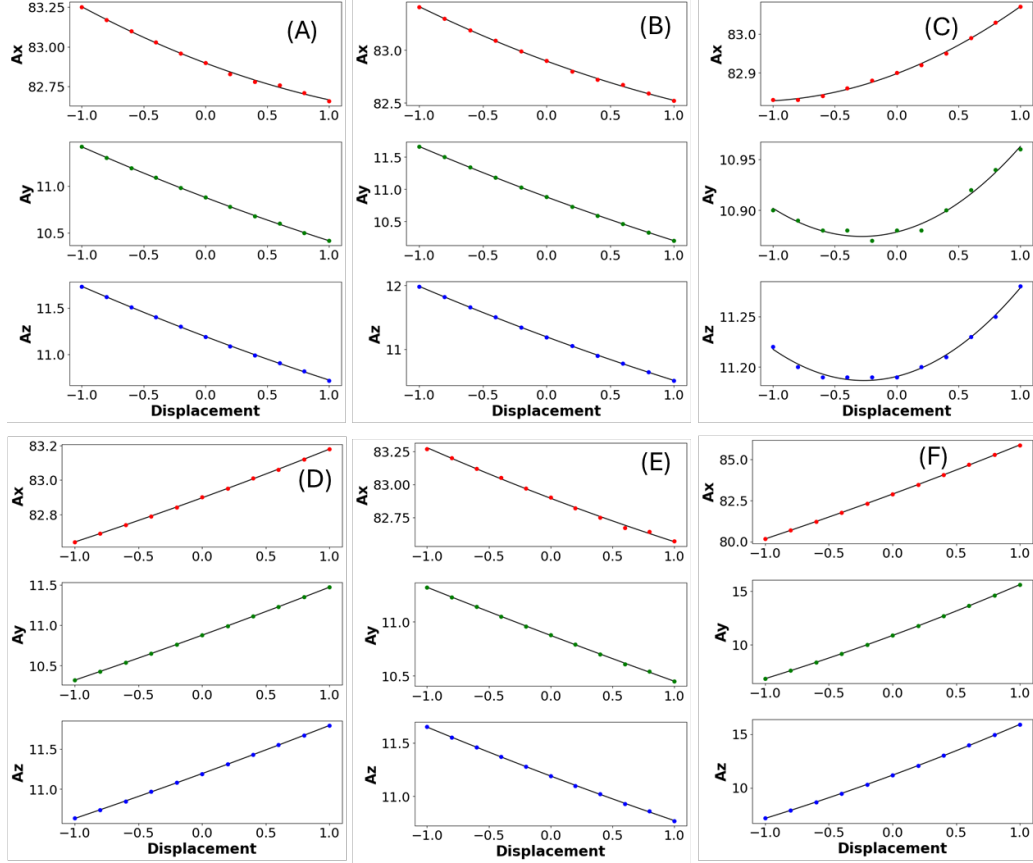


Figure 5: (A) to (F): Fitting of A matrix components to obtain coupling parameters for modes 1 to 6, respectively

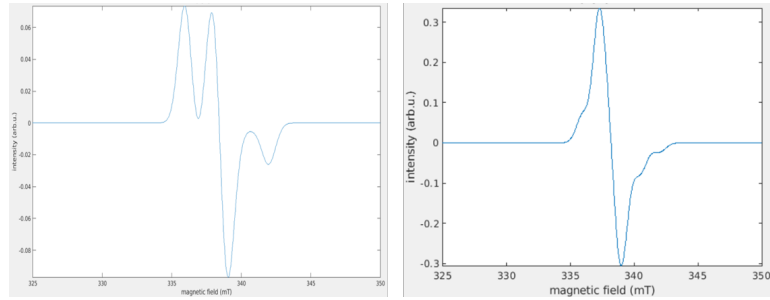


Figure 6: Left: Single TEMPO radical EPR
Right: Two TEMPO radicals, separated by 5 Å

5.5 TEMPO SAM

Now that we have analyzed the normal modes associated with a single spin centre, we extend our study to a self-assembled monolayer (SAM) of TEMPO radicals, in which multiple spin centres interact with one another while being anchored to a gold surface through alkyl-based linker units [3], starting from 11 carbon (Refer to Figure 7). In such a system, intermolecular interactions introduce an explicit spin–spin coupling term in the Hamiltonian, which was absent in the isolated molecule case. Our objective is to examine how variations in the alkyl chain length influence the decoherence behaviour

of these surface-bound radicals. We plan to employ *ab initio* molecular dynamics simulations in ORCA to gain deeper insight.

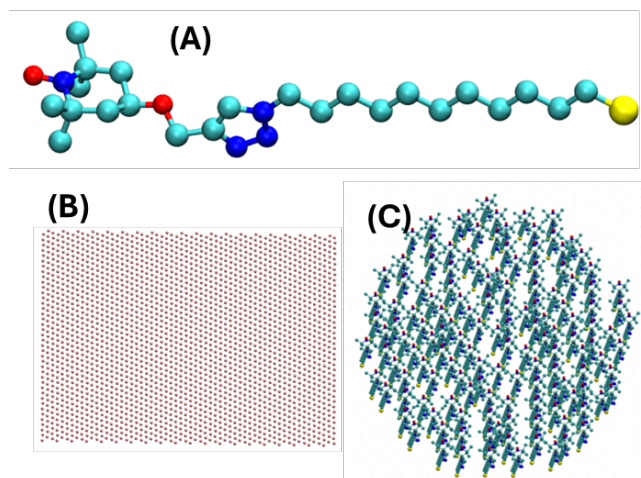


Figure 7: (A): TEMPO-OMe attached with linker, (B): Gold surface (111) plane, (C): Functionalized surface

6 Conclusion & Future Scope

This study has examined the structural and magnetic properties of TEMPO-based radicals using DFT, ORCA EPR calculations, and EasySpin simulations. Simulated EPR spectra reproduced the characteristic features of TEMPO and highlighted the effects of interactions between multiple radicals. We can also conclude that thermal occupation alone is therefore not a reliable indicator of the importance of a vibrational mode. The decisive factor is whether the mode involves geometrical changes that significantly affect the local environment of the unpaired electron.

Future work will focus on determining the relaxation times more directly, especially by combining *ab initio* molecular dynamics with EPR parameter calculations. Studying TEMPO units within self-assembled monolayers and systematically varying the alkyl-chain linker length will help clarify how structural flexibility affects decoherence. Such insights will guide the design of improved molecular qubits with enhanced coherence performance.

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